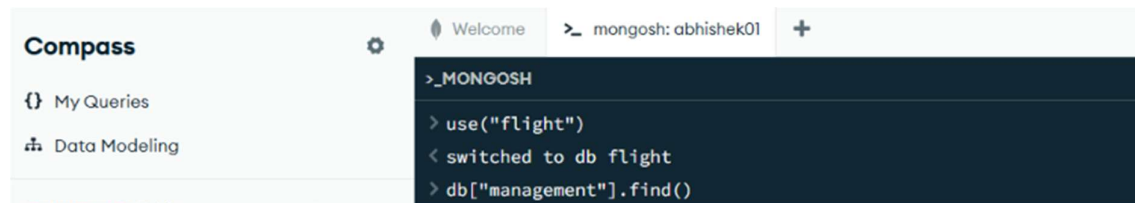


1.CREATE DATABASE

MongoDB Query:

Use Student

Explanation: Creates or switches to the Student database.

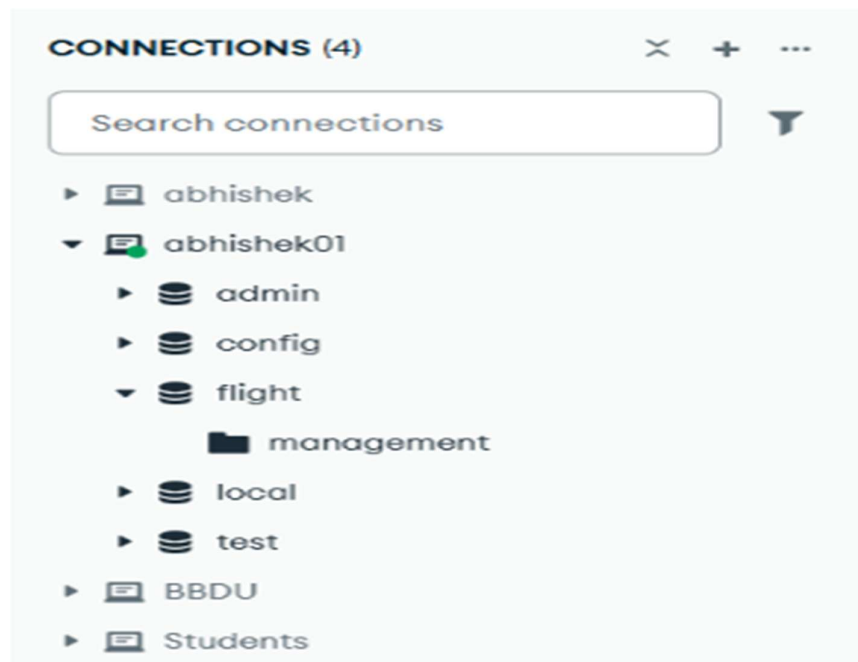


2. CREATE COLLECTION

MongoDB Query:

`db.create Collection("flight")`

Explanation: Creates a collection named flight.



3. INSERT FLIGHT DETAILS

MongoDB Query:

```
db.flights.insertMany([
  {
    flightId: "FL001",
    airline: "IndiGo",
    flightNumber: "6E201",
    aircraft: "Airbus A320",
    route: { from: "Delhi", to: "Mumbai", distanceKm: 1150 },
    departure: "06:30",
    arrival: "08:40",
    passengers: 164,
    seatsAvailable: 16,
    fare: 5200,
    baggageLimitKg: 15,
    status: "On Time"
  },
  {
    flightId: "FL002",
    airline: "Air India",
    flightNumber: "AI452",
    aircraft: "Airbus A321",
    route: { from: "Mumbai", to: "Bengaluru", distanceKm: 845 },
    departure: "08:15",
    arrival: "09:55",
    passengers: 172,
    seatsAvailable: 8,
```

```
fare: 6100,  
baggageLimitKg: 20,  
status: "On Time"  
},  
{  
flightId: "FL003",  
airline: "Vistara",  
flightNumber: "UK831",  
aircraft: "Boeing 737",  
route: { from: "Delhi", to: "Bengaluru", distanceKm: 1740 },  
departure: "10:00",  
arrival: "12:45",  
passengers: 138,  
seatsAvailable: 42,  
fare: 7200,  
baggageLimitKg: 20,  
status: "Delayed"  
},  
{  
flightId: "FL004",  
airline: "SpiceJet",  
flightNumber: "SG312",  
aircraft: "Boeing 737",  
route: { from: "Kolkata", to: "Delhi", distanceKm: 1300 },  
departure: "11:20",  
arrival: "13:40",  
passengers: 151,  
seatsAvailable: 29,
```

```
fare: 4800,  
baggageLimitKg: 15,  
status: "On Time"  
},  
{  
flightId: "FL005",  
airline: "IndiGo",  
flightNumber: "6E515",  
aircraft: "Airbus A320",  
route: { from: "Lucknow", to: "Mumbai", distanceKm: 1190 },  
departure: "07:45",  
arrival: "09:55",  
passengers: 156,  
seatsAvailable: 24,  
fare: 5100,  
baggageLimitKg: 15,  
status: "Boarding"  
},  
{  
flightId: "FL006",  
airline: "Air India",  
flightNumber: "AI671",  
aircraft: "Boeing 787",  
route: { from: "Delhi", to: "Dubai", distanceKm: 2190 },  
departure: "14:30",  
arrival: "17:10",  
passengers: 238,  
seatsAvailable: 12,
```

```
fare: 18500,  
baggageLimitKg: 30,  
status: "On Time"  
},  
{  
flightId: "FL007",  
airline: "Akasa Air",  
flightNumber: "QP142",  
aircraft: "Boeing 737 MAX",  
route: { from: "Mumbai", to: "Goa", distanceKm: 435 },  
departure: "09:20",  
arrival: "10:30",  
passengers: 162,  
seatsAvailable: 14,  
fare: 3900,  
baggageLimitKg: 15,  
status: "On Time"  
},  
{  
flightId: "FL008",  
airline: "IndiGo",  
flightNumber: "6E729",  
aircraft: "Airbus A321",  
route: { from: "Chennai", to: "Delhi", distanceKm: 1760 },  
departure: "16:10",  
arrival: "18:55",  
passengers: 181,  
seatsAvailable: 7,
```

```
fare: 6900,  
baggageLimitKg: 15,  
status: "Delayed"  
},  
{  
flightId: "FL009",  
airline: "Vistara",  
flightNumber: "UK944",  
aircraft: "Airbus A320",  
route: { from: "Hyderabad", to: "Delhi", distanceKm: 1250 },  
departure: "12:30",  
arrival: "14:40",  
passengers: 145,  
seatsAvailable: 35,  
fare: 5800,  
baggageLimitKg: 20,  
status: "On Time"  
},  
{  
flightId: "FL010",  
airline: "SpiceJet",  
flightNumber: "SG418",  
aircraft: "Boeing 737",  
route: { from: "Jaipur", to: "Mumbai", distanceKm: 920 },  
departure: "18:00",  
arrival: "19:45",  
passengers: 133,  
seatsAvailable: 51,
```

```
fare: 4500,  
baggageLimitKg: 15,  
status: "On Time"  
},
```

```
{  
  flightId: "FL011",  
  airline: "IndiGo",  
  flightNumber: "6E832",  
  aircraft: "Airbus A320",  
  route: { from: "Pune", to: "Delhi", distanceKm: 1175 },  
  departure: "05:50",  
  arrival: "07:55",  
  passengers: 158,  
  seatsAvailable: 22,  
  fare: 5300,  
  baggageLimitKg: 15,  
  status: "On Time"  
},
```

```
{  
  flightId: "FL012",  
  airline: "Air India",  
  flightNumber: "AI221",  
  aircraft: "Airbus A350",  
  route: { from: "Mumbai", to: "London", distanceKm: 7200 },  
  departure: "21:30",  
  arrival: "02:45",  
  passengers: 286,
```

```
seatsAvailable: 14,
fare: 52000,
baggageLimitKg: 30,
status: "On Time"
},
{
  flightId: "FL013",
  airline: "Akasa Air",
  flightNumber: "QP306",
  aircraft: "Boeing 737 MAX",
  route: { from: "Ahmedabad", to: "Delhi", distanceKm: 775 },
  departure: "13:15",
  arrival: "14:40",
  passengers: 149,
  seatsAvailable: 27,
  fare: 4200,
  baggageLimitKg: 15,
  status: "Boarding"
},
{
  flightId: "FL014",
  airline: "IndiGo",
  flightNumber: "6E409",
  aircraft: "Airbus A321",
  route: { from: "Kochi", to: "Bengaluru", distanceKm: 365 },
  departure: "06:20",
  arrival: "07:25",
  passengers: 174,
```

```
seatsAvailable: 10,  
fare: 3500,  
baggageLimitKg: 15,  
status: "On Time"  
},  
{  
flightId: "FL015",  
airline: "Vistara",  
flightNumber: "UK502",  
aircraft: "Airbus A320",  
route: { from: "Delhi", to: "Paris", distanceKm: 6570 },  
departure: "23:15",  
arrival: "05:40",  
passengers: 210,  
seatsAvailable: 18,  
fare: 47000,  
baggageLimitKg: 30,  
status: "On Time"  
},  
{  
flightId: "FL016",  
airline: "SpiceJet",  
flightNumber: "SG721",  
aircraft: "Boeing 737",  
route: { from: "Lucknow", to: "Delhi", distanceKm: 420 },  
departure: "09:10",  
arrival: "10:15",  
passengers: 142,
```

```
seatsAvailable: 38,  
fare: 3600,  
baggageLimitKg: 15,  
status: "On Time"  
},  
{  
flightId: "FL017",  
airline: "IndiGo",  
flightNumber: "6E611",  
aircraft: "Airbus A320",  
route: { from: "Delhi", to: "Kolkata", distanceKm: 1300 },  
departure: "15:20",  
arrival: "17:25",  
passengers: 167,  
seatsAvailable: 13,  
fare: 5700,  
baggageLimitKg: 15,  
status: "Delayed"  
},  
{  
flightId: "FL018",  
airline: "Air India",  
flightNumber: "AI508",  
aircraft: "Airbus A321",  
route: { from: "Chennai", to: "Mumbai", distanceKm: 1030 },  
departure: "10:45",  
arrival: "12:35",  
passengers: 155,
```

```
seatsAvailable: 25,  
fare: 5900,  
baggageLimitKg: 20,  
status: "On Time"  
},  
{  
  flightId: "FL019",  
  airline: "Akasa Air",  
  flightNumber: "QP711",  
  aircraft: "Boeing 737 MAX",  
  route: { from: "Bengaluru", to: "Kolkata", distanceKm: 1560 },  
  departure: "17:30",  
  arrival: "20:00",  
  passengers: 148,  
  seatsAvailable: 32,  
  fare: 6200,  
  baggageLimitKg: 15,  
  status: "On Time"  
},  
{  
  flightId: "FL020",  
  airline: "Vistara",  
  flightNumber: "UK715",  
  aircraft: "Airbus A320",  
  route: { from: "Mumbai", to: "Singapore", distanceKm: 3920 },  
  departure: "22:00",  
  arrival: "05:30",  
  passengers: 172,
```

```
seatsAvailable: 24,  
fare: 28500,  
baggageLimitKg: 25,  
status: "On Time"  
},
```

```
{  
  flightId: "FL021",  
  airline: "IndiGo",  
  flightNumber: "6E923",  
  aircraft: "Airbus A320",  
  route: { from: "Delhi", to: "Varanasi", distanceKm: 680 },  
  departure: "07:00",  
  arrival: "08:15",  
  passengers: 151,  
  seatsAvailable: 29,  
  fare: 4100,  
  baggageLimitKg: 15,  
  status: "On Time"  
},
```

```
{  
  flightId: "FL022",  
  airline: "Air India",  
  flightNumber: "AI734",  
  aircraft: "Airbus A321",  
  route: { from: "Bengaluru", to: "Hyderabad", distanceKm: 500 },  
  departure: "11:30",  
  arrival: "12:35",
```

```
passengers: 160,
seatsAvailable: 20,
fare: 3900,
baggageLimitKg: 20,
status: "Boarding"
},
{
  flightId: "FL023",
  airline: "SpiceJet",
  flightNumber: "SG829",
  aircraft: "Boeing 737",
  route: { from: "Delhi", to: "Goa", distanceKm: 1510 },
  departure: "13:45",
  arrival: "16:20",
  passengers: 139,
  seatsAvailable: 41,
  fare: 6100,
  baggageLimitKg: 15,
  status: "On Time"
},
{
  flightId: "FL024",
  airline: "Akasa Air",
  flightNumber: "QP418",
  aircraft: "Boeing 737 MAX",
  route: { from: "Mumbai", to: "Delhi", distanceKm: 1150 },
  departure: "19:10",
  arrival: "21:20",
```

```
passengers: 169,  
seatsAvailable: 11,  
fare: 5600,  
baggageLimitKg: 15,  
status: "Delayed"  
},  
{  
  flightId: "FL025",  
  airline: "Vistara",  
  flightNumber: "UK623",  
  aircraft: "Airbus A321",  
  route: { from: "Delhi", to: "Hyderabad", distanceKm: 1260 },  
  departure: "08:50",  
  arrival: "10:50",  
  passengers: 177,  
  seatsAvailable: 3,  
  fare: 6500,  
  baggageLimitKg: 20,  
  status: "On Time"  
},  
{  
  flightId: "FL026",  
  airline: "IndiGo",  
  flightNumber: "6E302",  
  aircraft: "Airbus A320",  
  route: { from: "Kolkata", to: "Bengaluru", distanceKm: 1560 },  
  departure: "14:20",  
  arrival: "16:55",
```

```
passengers: 163,
seatsAvailable: 17,
fare: 6400,
baggageLimitKg: 15,
status: "On Time"
},
{
  flightId: "FL027",
  airline: "Air India",
  flightNumber: "AI902",
  aircraft: "Boeing 787",
  route: { from: "Delhi", to: "New York", distanceKm: 11700 },
  departure: "01:10",
  arrival: "07:20",
  passengers: 241,
  seatsAvailable: 17,
  fare: 72000,
  baggageLimitKg: 30,
  status: "On Time"
},
{
  flightId: "FL028",
  airline: "SpiceJet",
  flightNumber: "SG533",
  aircraft: "Boeing 737",
  route: { from: "Mumbai", to: "Jaipur", distanceKm: 920 },
  departure: "12:15",
  arrival: "14:00",
```

```
passengers: 127,  
seatsAvailable: 57,  
fare: 4300,  
baggageLimitKg: 15,  
status: "On Time"  
},  
{  
  flightId: "FL029",  
  airline: "IndiGo",  
  flightNumber: "6E801",  
  aircraft: "Airbus A321",  
  route: { from: "Lucknow", to: "Bengaluru", distanceKm: 1580 },  
  departure: "16:30",  
  arrival: "19:10",  
  passengers: 175,  
  seatsAvailable: 9,  
  fare: 6800,  
  baggageLimitKg: 15,  
  status: "Boarding"  
},  
{  
  flightId: "FL030",  
  airline: "Vistara",  
  flightNumber: "UK910",  
  aircraft: "Airbus A320",  
  route: { from: "Chennai", to: "Delhi", distanceKm: 1760 },  
  departure: "20:30",  
  arrival: "23:15",
```

```
passengers: 143,  
seatsAvailable: 37,  
fare: 7100,  
baggageLimitKg: 20,  
status: "On Time"  
},
```

```
{  
  flightId: "FL031",  
  airline: "Akasa Air",  
  flightNumber: "QP521",  
  aircraft: "Boeing 737 MAX",  
  route: { from: "Pune", to: "Bengaluru", distanceKm: 725 },  
  departure: "07:30",  
  arrival: "08:45",  
  passengers: 154,  
  seatsAvailable: 26,  
  fare: 4400,  
  baggageLimitKg: 15,  
  status: "On Time"  
},
```

```
{  
  flightId: "FL032",  
  airline: "Air India",  
  flightNumber: "AI310",  
  aircraft: "Airbus A350",  
  route: { from: "Mumbai", to: "Tokyo", distanceKm: 6730 },  
  departure: "18:45",
```

```
    arrival: "07:15",
    passengers: 275,
    seatsAvailable: 25,
    fare: 58000,
    baggageLimitKg: 30,
    status: "On Time"
  },
  {
    flightId: "FL033",
    airline: "IndiGo",
    flightNumber: "6E440",
    aircraft: "Airbus A320",
    route: { from: "Delhi", to: "Ahmedabad", distanceKm: 775 },
    departure: "09:40",
    arrival: "11:05",
    passengers: 159,
    seatsAvailable: 21,
    fare: 4500,
    baggageLimitKg: 15,
    status: "On Time"
  },
  {
    flightId: "FL034",
    airline: "SpiceJet",
    flightNumber: "SG615",
    aircraft: "Boeing 737",
    route: { from: "Kolkata", to: "Mumbai", distanceKm: 1650 },
    departure: "15:50",
```

```
    arrival: "18:35",
    passengers: 146,
    seatsAvailable: 34,
    fare: 6700,
    baggageLimitKg: 15,
    status: "Delayed"
  },
  {
    flightId: "FL035",
    airline: "Vistara",
    flightNumber: "UK802",
    aircraft: "Airbus A321",
    route: { from: "Delhi", to: "Dubai", distanceKm: 2190 },
    departure: "20:15",
    arrival: "23:00",
    passengers: 183,
    seatsAvailable: 17,
    fare: 19200,
    baggageLimitKg: 25,
    status: "On Time"
  },
  {
    flightId: "FL036",
    airline: "IndiGo",
    flightNumber: "6E556",
    aircraft: "Airbus A320",
    route: { from: "Hyderabad", to: "Goa", distanceKm: 540 },
    departure: "06:45",
```

```
    arrival: "07:55",
    passengers: 148,
    seatsAvailable: 32,
    fare: 3800,
    baggageLimitKg: 15,
    status: "On Time"
  },
  {
    flightId: "FL037",
    airline: "Air India",
    flightNumber: "AI512",
    aircraft: "Airbus A321",
    route: { from: "Lucknow", to: "Mumbai", distanceKm: 1190 },
    departure: "12:40",
    arrival: "14:50",
    passengers: 161,
    seatsAvailable: 19,
    fare: 5700,
    baggageLimitKg: 20,
    status: "Boarding"
  },
  {
    flightId: "FL038",
    airline: "Akasa Air",
    flightNumber: "QP631",
    aircraft: "Boeing 737 MAX",
    route: { from: "Delhi", to: "Pune", distanceKm: 1175 },
    departure: "17:00",
```

```
    arrival: "19:05",
    passengers: 153,
    seatsAvailable: 27,
    fare: 5300,
    baggageLimitKg: 15,
    status: "On Time"
  },
  {
    flightId: "FL039",
    airline: "Vistara",
    flightNumber: "UK731",
    aircraft: "Airbus A320",
    route: { from: "Bengaluru", to: "Mumbai", distanceKm: 845 },
    departure: "10:15",
    arrival: "11:50",
    passengers: 170,
    seatsAvailable: 10,
    fare: 5800,
    baggageLimitKg: 20,
    status: "On Time"
  },
  {
    flightId: "FL040",
    airline: "SpiceJet",
    flightNumber: "SG745",
    aircraft: "Boeing 737",
    route: { from: "Jaipur", to: "Delhi", distanceKm: 240 },
    departure: "18:30",
```

```
    arrival: "19:20",  
    passengers: 135,  
    seatsAvailable: 45,  
    fare: 3200,  
    baggageLimitKg: 15,  
    status: "On Time"  
  },
```

```
{  
  flightId: "FL041",  
  airline: "IndiGo",  
  flightNumber: "6E911",  
  aircraft: "Airbus A321",  
  route: { from: "Delhi", to: "Srinagar", distanceKm: 650 },  
  departure: "06:15",  
  arrival: "07:35",  
  passengers: 171,  
  seatsAvailable: 9,  
  fare: 4900,  
  baggageLimitKg: 15,  
  status: "On Time"  
},
```

```
{  
  flightId: "FL042",  
  airline: "Air India",  
  flightNumber: "AI638",  
  aircraft: "Airbus A320",  
  route: { from: "Mumbai", to: "Ahmedabad", distanceKm: 440 },
```

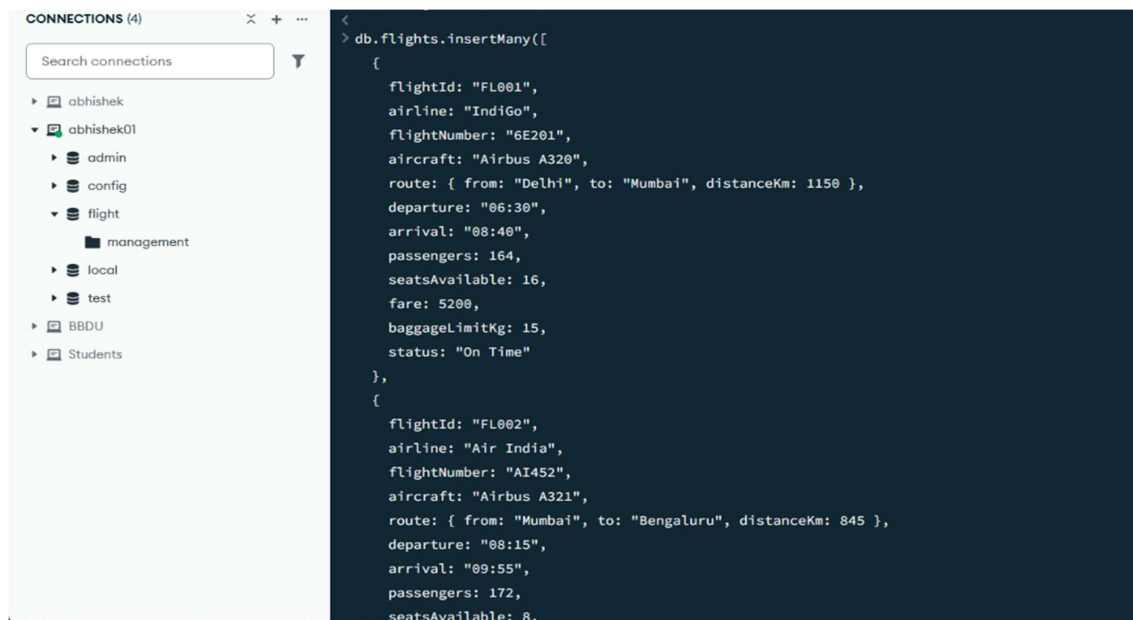
```
departure: "09:00",
arrival: "10:00",
passengers: 147,
seatsAvailable: 33,
fare: 3600,
baggageLimitKg: 20,
status: "On Time"
},
{
  flightId: "FL043",
  airline: "Akasa Air",
  flightNumber: "QP752",
  aircraft: "Boeing 737 MAX",
  route: { from: "Kolkata", to: "Delhi", distanceKm: 1300 },
  departure: "13:10",
  arrival: "15:25",
  passengers: 152,
  seatsAvailable: 28,
  fare: 5200,
  baggageLimitKg: 15,
  status: "Delayed"
},
{
  flightId: "FL044",
  airline: "Vistara",
  flightNumber: "UK845",
  aircraft: "Airbus A321",
  route: { from: "Delhi", to: "Singapore", distanceKm: 4150 },
```

```
    departure: "22:30",
    arrival: "06:50",
    passengers: 185,
    seatsAvailable: 15,
    fare: 31000,
    baggageLimitKg: 25,
    status: "On Time"
  },
  {
    flightId: "FL045",
    airline: "SpiceJet",
    flightNumber: "SG912",
    aircraft: "Boeing 737",
    route: { from: "Lucknow", to: "Kolkata", distanceKm: 870 },
    departure: "08:25",
    arrival: "10:10",
    passengers: 129,
    seatsAvailable: 51,
    fare: 4700,
    baggageLimitKg: 15,
    status: "On Time"
  },
  {
    flightId: "FL046",
    airline: "IndiGo",
    flightNumber: "6E677",
    aircraft: "Airbus A320",
    route: { from: "Mumbai", to: "Kochi", distanceKm: 1060 },
```

```
departure: "14:45",
arrival: "16:35",
passengers: 166,
seatsAvailable: 14,
fare: 5500,
baggageLimitKg: 15,
status: "Boarding"
},
{
  flightId: "FL047",
  airline: "Air India",
  flightNumber: "AI822",
  aircraft: "Airbus A321",
  route: { from: "Delhi", to: "Kochi", distanceKm: 2080 },
  departure: "19:45",
  arrival: "22:50",
  passengers: 173,
  seatsAvailable: 7,
  fare: 7600,
  baggageLimitKg: 20,
  status: "On Time"
},
{
  flightId: "FL048",
  airline: "Akasa Air",
  flightNumber: "QP863",
  aircraft: "Boeing 737 MAX",
  route: { from: "Bengaluru", to: "Pune", distanceKm: 725 },
```

```
    departure: "11:15",
    arrival: "12:35",
    passengers: 141,
    seatsAvailable: 39,
    fare: 4200,
    baggageLimitKg: 15,
    status: "On Time"
  },
  {
    flightId: "FL049",
    airline: "Vistara",
    flightNumber: "UK956",
    aircraft: "Airbus A320",
    route: { from: "Mumbai", to: "Delhi", distanceKm: 1150 },
    departure: "16:45",
    arrival: "18:50",
    passengers: 179,
    seatsAvailable: 1,
    fare: 6200,
    baggageLimitKg: 20,
    status: "On Time"
  },
  {
    flightId: "FL050",
    airline: "IndiGo",
    flightNumber: "6E999",
    aircraft: "Airbus A321",
    route: { from: "Delhi", to: "Goa", distanceKm: 1510 },
```

```
departure: "21:00",  
arrival: "23:35",  
passengers: 176,  
seatsAvailable: 8,  
fare: 6400,  
baggageLimitKg: 15,  
status: "On Time"  
}  
])
```



Explanation: Inserts multiple student documents into the collection

4. DISPLAY ALL FLIGHT

MongoDB Query:

```
db.flights.find()
```

Explanation: Retrieves documents from the Flight collection.

```
> db.flights.find()
{
  _id: ObjectId('6ab94be25a38ce94938c59c1'),
  flightId: 'FL001',
  airline: 'IndiGo',
  flightNumber: '6E201',
  aircraft: 'Airbus A320',
  route: {
    from: 'Delhi',
    to: 'Mumbai',
    distanceKm: 1150
  },
  departure: '06:30',
  arrival: '08:40',
  passengers: 164,
  seatsAvailable: 16,
  fare: 5200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

5. Search Flights by Departure City

MongoDB Query:

```
db.flights.find({"route.from": "Mumbai"})
```

Explanation: This operation finds all flights departing from Mumbai.

```
> db.flights.find({"route.from": "Mumbai"})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c2'),
  flightId: 'FL002',
  airline: 'Air India',
  flightNumber: 'AI452',
  aircraft: 'Airbus A321',
  route: {
    from: 'Mumbai',
    to: 'Bengaluru',
    distanceKm: 845
  },
  departure: '08:15',
  arrival: '09:55',
  passengers: 172,
  seatsAvailable: 8,
  fare: 6100,
  baggageLimitKg: 20,
  status: 'On Time'
}
```

6. Find Flights with Fare Greater Than ₹6000

MongoDB Query:

```
db.flights.find({fare: {$gt: 6000}})
```

Explanation: This operation finds flights whose fare is greater than ₹6000.

```
> db.flights.find({fare: {$gt: 6000}})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c2'),
  flightId: 'FL002',
  airline: 'Air India',
  flightNumber: 'AI452',
  aircraft: 'Airbus A321',
  route: {
    from: 'Mumbai',
    to: 'Bengaluru',
    distanceKm: 845
  },
  departure: '08:15',
  arrival: '09:55',
  passengers: 172,
  seatsAvailable: 8,
  fare: 6100,
  baggageLimitKg: 20,
  status: 'On Time'
}
```

7. Find Flights with More Than 150 Passengers

MongoDB Query:

```
db.flights.find({passengers: {$gt: 150}})
```

Purpose: This operation finds flights having more than 150 passengers.

```
> db.flights.find({passengers: {$gt: 150}})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c1'),
  flightId: 'FL001',
  airline: 'IndiGo',
  flightNumber: '6E201',
  aircraft: 'Airbus A320',
  route: {
    from: 'Delhi',
    to: 'Mumbai',
    distanceKm: 1150
  },
  departure: '06:30',
  arrival: '08:40',
  passengers: 164,
  seatsAvailable: 16,
  fare: 5200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

8. Search Delayed Flights

MongoDB Query:

```
db.flights.find({status: "Delayed"})
```

Explanation: This operation displays all flights whose current status is delayed.

```
> db.flights.find({status: "Delayed"})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c3'),
  flightId: 'FL003',
  airline: 'Vistara',
  flightNumber: 'UK831',
  aircraft: 'Boeing 737',
  route: {
    from: 'Delhi',
    to: 'Bengaluru',
    distanceKm: 1740
  },
  departure: '10:00',
  arrival: '12:45',
  passengers: 138,
  seatsAvailable: 42,
  fare: 7200,
  baggageLimitKg: 20,
  status: 'Delayed'
}
```

9. Search Flights by Source and Destination

MongoDB Query:

```
db.flights.find({status: "Delayed"})
```

Explanation: This operation finds flights travelling from Delhi to Mumbai.

```
> db.flights.find({
  "route.from": "Delhi",
  "route.to": "Mumbai"
})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c1'),
  flightId: 'FL001',
  airline: 'IndiGo',
  flightNumber: '6E201',
  aircraft: 'Airbus A320',
  route: {
    from: 'Delhi',
    to: 'Mumbai',
    distanceKm: 1150
  },
  departure: '06:30',
  arrival: '08:40',
  passengers: 164,
  seatsAvailable: 16,
  fare: 5200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

10. Sort Flights by Fare in Descending Order

MongoDB Query:

```
db.flights.find().sort({fare: -1})
```

Purpose: This operation displays flights from the highest fare to the lowest fare.

```
> db.flights.find().sort({fare: -1})
< {
  _id: ObjectId('6ab94be25a38ce94938c59db'),
  flightId: 'FL027',
  airline: 'Air India',
  flightNumber: 'AI902',
  aircraft: 'Boeing 787',
  route: {
    from: 'Delhi',
    to: 'New York',
    distanceKm: 11700
  },
  departure: '01:10',
  arrival: '07:20',
  passengers: 241,
  seatsAvailable: 17,
  fare: 72000,
  baggageLimitKg: 30,
  status: 'On Time'
}
```

11. Sort Flights by Fare in Ascending Order

MongoDB Query:

```
db.flights.find().sort({fare: 1})
```

Explanation: This operation displays flights from the lowest fare to the highest fare.

```
> db.flights.find().sort({fare: 1})
< {
  _id: ObjectId('6ab94be25a38ce94938c59e8'),
  flightId: 'FL040',
  airline: 'SpiceJet',
  flightNumber: 'SG745',
  aircraft: 'Boeing 737',
  route: {
    from: 'Jaipur',
    to: 'Delhi',
    distanceKm: 240
  },
  departure: '18:30',
  arrival: '19:20',
  passengers: 135,
  seatsAvailable: 45,
  fare: 3200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

11. Count Total Number of Flights

MongoDB Query:

```
db.flights.countDocuments()
```

Explanation: **This operation counts the total number of flight records stored in the collection.**

```
> db.flights.countDocuments()  
< 50
```

13. Search Flights by Aircraft Type Using Regex

MongoDB Query:

```
db.flights.find({ aircraft: {$regex: "Airbus"}})
```

Explanation: *This operation searches for flights that use Airbus aircraft.*

```
> db.flights.find({aircraft: {$regex: "Airbus"}})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c1'),
  flightId: 'FL001',
  airline: 'IndiGo',
  flightNumber: '6E201',
  aircraft: 'Airbus A320',
  route: {
    from: 'Delhi',
    to: 'Mumbai',
    distanceKm: 1150
  },
  departure: '06:30',
  arrival: '08:40',
  passengers: 164,
  seatsAvailable: 16,
  fare: 5200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

14. Perform AND Operation on Flight Records

MongoDB Query:

```
db.flights.find({ $and: [ {passengers: {$gt: 150}}, {fare: {$lt: 7000}} ] })
```

Explanation: This operation finds flights having more than 150 passengers AND a fare below ₹7000.

```
> db.flights.find({
  $and: [
    {passengers: {$gt: 150}},
    {fare: {$lt: 7000}}
  ]
})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c1'),
  flightId: 'FL001',
  airline: 'IndiGo',
  flightNumber: '6E201',
  aircraft: 'Airbus A320',
  route: {
    from: 'Delhi',
    to: 'Mumbai',
    distanceKm: 1150
  },
  departure: '06:30',
  arrival: '08:40',
  passengers: 164,
  seatsAvailable: 16,
  fare: 5200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

15. Perform OR Operation on Flight Records

A.MongoDB Query:

```
db.flights.find({ $or: [ {status: "Delayed"}, {seatsAvailable: {$lt: 10}} ] })
```

Explanation: This operation finds flights that are either delayed OR have fewer than 10 available seats.

```
> db.flights.find({
  $or: [
    {status: "Delayed"},
    {seatsAvailable: {$lt: 10}}
  ]
})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c2'),
  flightId: 'FL002',
  airline: 'Air India',
  flightNumber: 'AI452',
  aircraft: 'Airbus A321',
  route: {
    from: 'Mumbai',
    to: 'Bengaluru',
    distanceKm: 845
  },
  departure: '08:15',
  arrival: '09:55',
  passengers: 172,
  seatsAvailable: 8,
  fare: 6100,
  baggageLimitKg: 20,
  status: 'On Time'
}
```

16. Find Flights with Available Seats Less Than 20

MongoDB Query:

```
db.flights.find({ seatsAvailable: {$lt: 20} })
```

Explanation: This operation finds flights having fewer than 20 available seats.

```
> db.flights.find({
  seatsAvailable: {$lt: 20}
})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c1'),
  flightId: 'FL001',
  airline: 'IndiGo',
  flightNumber: '6E201',
  aircraft: 'Airbus A320',
  route: {
    from: 'Delhi',
    to: 'Mumbai',
    distanceKm: 1150
  },
  departure: '06:30',
  arrival: '08:40',
  passengers: 164,
  seatsAvailable: 16,
  fare: 5200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

17. Find Flights with Fare Between ₹4000 and ₹7000

MongoDB Query:

```
db.flights.find({ fare: {$gte: 4000, $lte: 7000} })
```

Explanation: This operation finds flights whose fare is between ₹4000 and ₹7000.

```
> db.flights.find({
  fare: {$gte: 4000, $lte: 7000}
})
< {
  _id: ObjectId('6ab94be25a38ce94938c59c1'),
  flightId: 'FL001',
  airline: 'IndiGo',
  flightNumber: '6E201',
  aircraft: 'Airbus A320',
  route: {
    from: 'Delhi',
    to: 'Mumbai',
    distanceKm: 1150
  },
  departure: '06:30',
  arrival: '08:40',
  passengers: 164,
  seatsAvailable: 16,
  fare: 5200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

18. Find Flights by Airline

MongoDB Query:

```
db.flights.find({ airline: "IndiGo" })
```

Explanation: This operation finds all flights operated by IndiGo.

```
> db.flights.find({ airline: "IndiGo" })
< {
  _id: ObjectId('6ab94be25a38ce94938c59c1'),
  flightId: 'FL001',
  airline: 'IndiGo',
  flightNumber: '6E201',
  aircraft: 'Airbus A320',
  route: {
    from: 'Delhi',
    to: 'Mumbai',
    distanceKm: 1150
  },
  departure: '06:30',
  arrival: '08:40',
  passengers: 164,
  seatsAvailable: 16,
  fare: 5200,
  baggageLimitKg: 15,
  status: 'On Time'
}
```

MONGODB OPERATION USED:

1. Create Database & Collection
2. Insert Operation — insertMany()
3. Find / Display Operation — find()
4. Search Operation — find()
5. Comparison Operators — \$gt, \$lt, \$gte, \$lte
6. Sort Operation — sort()
 - Ascending 1
 - Descending -1
7. Count Operation — countDocuments()
8. Regex Operation — \$regex
9. AND Operation — \$and
10. OR Operation — \$or
11. Nested/Embedded Document Search —
"route.from"
12. Multiple Condition Filtering — multiple
conditions in one query

CONCLUSION

The Flight Management Records project demonstrates how MongoDB can be used to store and manage flight data efficiently. Various MongoDB operations such as Insert, Find, Search, Sort, Count, Regex, AND, OR, and comparison operators were performed on flight records. This project helped in understanding the practical use of NoSQL databases for managing real-world data in a flexible and efficient manner.